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Response to Brock: noise and autism

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As Brock [1] suggests, it is indeed difficult to distinguish the action of top-down from bottom-up processes within a Bayesian framework. Either reducing the variance (noisiness) of the likelihood (bottom-up) or increasing that of the prior (top-down) has the same net result in changing their relative weighting in determining the posterior. Either possibility is, therefore, consistent with our framework. Although we welcome discussion of the causal underpinnings of altered sensation and perception in autism, we believe that an entirely bottom-up account is both unsupported by the evidence and limited in its explanatory scope.

Multiple factors contribute to the internal neural noise that determines the variance of the likelihood of a visual judgment, including photon shot-noise, optical distortions, thermal photoreceptor noise, neural transmission and processing noise, and sampling noise from the necessarily limited neural resources. The common consensus (e.g., [2]) is that animal sensory systems have evolved to perform optimally in their environment, so there is little room for further noise reduction. Reduced noisiness in autistic perception would require sensory systems built with superior components (or perhaps more of them). This is not entirely impossible, but it is not obvious from the available evidence. Indeed, the evidence from imaging and electroencephalography (EEG) studies suggests that the brains of autistic people are characterized by more noise than those of non-autistic people (e.g., [3]; also see [4,5]). Similarly, there is little psychophysical evidence for reduced noise in autism, which should lead to more precise psychophysical judgments in basic tasks that are uninfluenced by priors (see [4] for review).

The Bayesian model is often considered optimal in the sense that the priors are adjusted dynamically to reduce overall error. If the likelihood were less noisy in autistic perception, a functional Bayesian integrator

would have to adjust the priors appropriately, in the same way that the priors of professional percussionists (who have enhanced temporal discrimination) are different from non-percussionists in discriminating temporal intervals [6], all leading to optimal performance. However, autistic perception does not always confer advantages [7], especially when prior knowledge is necessary to resolve perceptual ambiguities (e.g., [8]). To explain these data and to provide a more complete explanatory account of the perceptual experiences of autistic people (one that goes beyond the focus on enhanced, or hyper-, sensitivity in autism), Brock's account would require proposing reduced sensory noise in autism and atypicalities in the dynamic calculation of the prior, which would fail to take the reduced noisiness into account. Our concept of ‘hypo-priors’ [9] has greater explanatory power than existing bottom-up accounts, accounting not only for the reported hypersensitivity in autism, but also for a range of other sensory and other non-social symptoms.

Most importantly, contrary to Brock's suggestion, we believe that our account and the Bayesian framework more broadly has firm implications for understanding how autistic people represent information at all levels of the perceptual and cognitive system, including complex situations, such as social interactions. We chose, however, to begin our approach with the more tractable problem of perception, which is amenable to investigation with reasonably well-controlled stimuli, before extending these principles to the social realm.

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